

Blair & Rosen Inc. Brokerage firm

1 Problem

Building a Financial Portfolio. Blair & Rosen, Inc. (B&R) is a brokerage firm that specializes in investment portfolios designed to meet the specific risk tolerances of its clients. A client who contacted B&R this past week has a maximum of \$50,000 to invest. B&R's investment advisor decides to recommend a portfolio consisting of two investment funds: an Internet fund and a Blue Chip fund. The Internet fund has a projected annual return of 12%, and the Blue Chip fund has a projected annual return of 9%. The investment advisor requires that at most \$35,000 of the client's funds should be invested in the Internet fund. B&R services include a risk rating for each investment alternative. The Internet fund, which is the more risky of the two investment alternatives, has a risk rating of 6 per \$1,000 invested. The Blue Chip fund has a risk rating of 4 per \$1,000 invested. For example, if \$10,000 is invested in each of the two investment funds, B&R's risk rating for the portfolio would be $6(10) + 4(10) = 100$. Finally, B&R developed a questionnaire to measure each client's risk tolerance. Based on the responses, each client is classified as a conservative, moderate, or aggressive investor. Suppose that the questionnaire results classified the current client as a moderate investor. B&R recommends that a client who is a moderate investor limit his or her portfolio to a maximum risk rating of 240.

2 Data

	Internet Fund	Blue chip fund
Annual Return	12%	9%
Risk Rating (per \$1000)	6	4
Max investment Allowed	\$35.000	-

Let

$F = \{\text{Internet Fund}, \text{Blue chit fund}\}$

$return_i = \text{Annual returns from } i, i \in F$

$risk_i = \text{Risk rating for } i, i \in F$

3 Objective in Words

Decide **how much to invest in the Internet fund and the Blue Chip fund** so that **the total return (in \$)** is maximized, subject to the following constraints:

- **The total amount invested** must not exceed \$50,000.
- **No more than \$35,000** should be invested in the Internet fund.
- **The total risk rating** of the portfolio must not exceed 240.

4 Decision Variables

Let

x_1 = *The amount of money invested in the Internet Fund (in dollars)*

x_2 = *The amount of money invested in the Blue Chip Fund (in dollars).*

x_i = *The amount of money invested in i , $i \in F$*

5 Algebraic Formulation

$$\text{Maximize } \sum_{i \in F} \text{return}_i x_i$$

Such that

$$\sum_{i \in F} x_i \leq 50,000 \quad (\text{Total Investment Constraint})$$

$$x_1 \leq 35,000 \quad (\text{Internet Fund Constraint})$$

$$\sum_{i \in F} \text{risk}_i \cdot \frac{x_i}{1000} \leq 240 \quad (\text{Risk Constraint})$$

$$x_1, x_2 \geq 0 \quad (\text{Non - Negativity Constraint})$$

6 Implementation

An implementation and solution of the model using Python, AMPL is available here:

<https://colab.research.google.com/drive/17zsyYV17O9GG7VIwYuunoYRZKuWqXT0T?usp=sharing>

7 Results

The optimal solution is to invest **\$20,000 in the Internet Fund** and **\$30,000 in the Blue-Chip Fund** for a **total return of \$5,100**.

8 ChatGPT Prompt :

Coaching a generative AI (I have used ChatGPT) to produce AMPL code to solve the problem is shown here:

<https://chatgpt.com/share/66eb4008-4504-800a-9211-d7d7f2c744bc>